

ASSIGNMENT ON CSE-260

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Answer to the Question No. 1

According to the state diagram, we construct the state table which is:-

A	B	x	A^+	B^+
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Now for S-R flip Flop

A	B	x	A^+	B^+	$S(A)$	$R(A)$	$S(B)$	$R(B)$
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	0
0	1	0	1	0	1	0	0	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	0
1	1	0	1	1	X	0	X	0
1	1	1	0	0	0	1	0	1

Now we construct the K-Maps for the flip flop inputs:-

A \ Bx	Bx	B'x'	B'x	Bx	Bx'
A'	0	0	0	0	1
A	X	X	0	0	X

$$\therefore S(A) = Bx'$$

A \ Bx	Bx	B'x'	B'x	Bx	Bx'
A'	X	X	X	0	0
A	0	0	1	0	0

$$\therefore R(A) = Bx$$

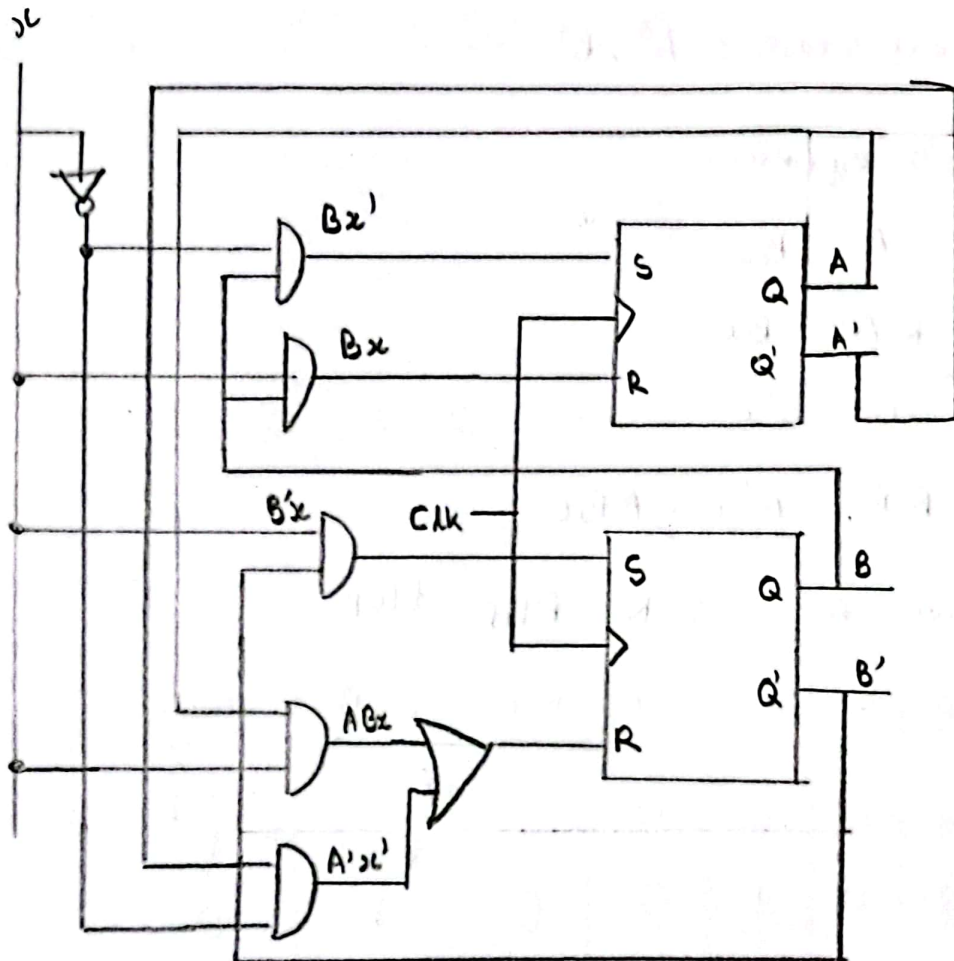
A \ Bx	Bx	B'x'	B'x	Bx	Bx'
A'	0	1	X	0	0
A	0	1	0	X	0

$$S(B) = B'x$$

A \ Bx	Bx	B'x'	B'x	Bx	Bx'
A'	X	0	0	1	0
A	X	0	1	0	0

$$R(B) = A'x' + ABx$$

Now we construct the circuit



Now,

Present state A, B

Inputs : S A, R A, S B, R B, x

Next state : A^+ , B^+

Now equations

$$S A = B x'$$

$$R A = B x$$

$$S B = B' x$$

$$R B = A' x' + A B x$$

Now for S-R Flip flop,

A	B	x	S(A)	R(A)	S(B)	R(B)
0	0	0	0	0	0	1
0	0	1	0	0	1	0
0	1	0	1	0	0	1
0	1	1	0	1	0	0
1	0	0	0	0	0	0
1	0	1	0	0	1	0
1	1	0	1	0	0	0
1	1	1	0	1	0	1

For the next state,

A	B	x	$S(A)$	$R(A)$	$S(B)$	$R(B)$	A'	B'
0	0	0	0	0	0	1	0	0
0	0	1	0	0	1	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	1	0	0	0	1
1	0	0	0	0	0	0	1	0
1	0	1	0	0	1	0	1	1
1	1	0	1	0	0	0	1	1
1	1	1	0	1	0	1	0	0

The required state diagram is :-

